Getting the Grade

Affordably Improving the Energy Efficiency and Carbon Emissions of Buildings in NYC

Project Team

Justin Seret

User experience researcher, human factors engineer, and educator with experience across the medical technology sector.

Sara Pardee

Recent graduate from Johns Hopkins with a focus on mechanical and environmental engineering.

Sofia Fernandez

Software engineer with experience in big tech, looking to transition into climate tech

Rod Mobini

Engineering leader dedicated to using teamwork, science, and technology.

Marina Rayman

Senior strategist with diverse financial services experience across public and private asset classes, product advocacy, business management, technology and senior stakeholder engagement.

Cecilia Cheng

Operations strategist with consulting experience for financial service organizations, project management and finance operations

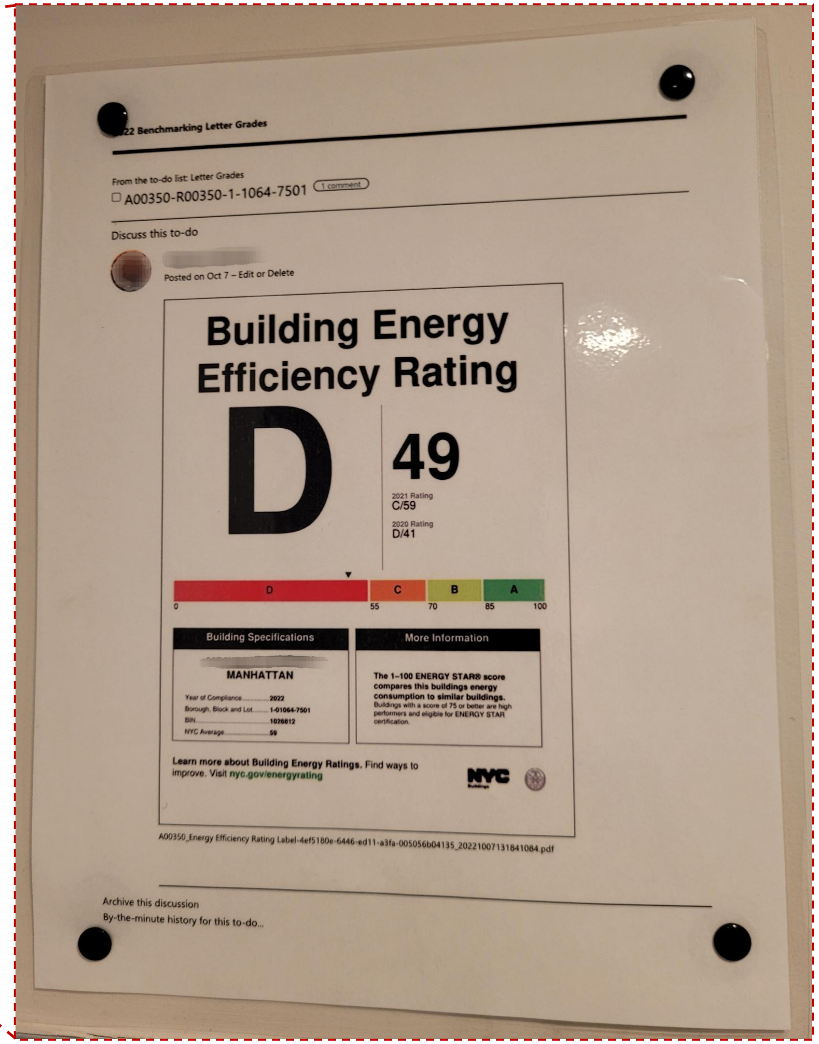
Yujia Han

Researcher specializing in sustainable finance, energy policy, and development economics.

Theo Manton

Frontend-leaning software engineer with a strong sense of design.

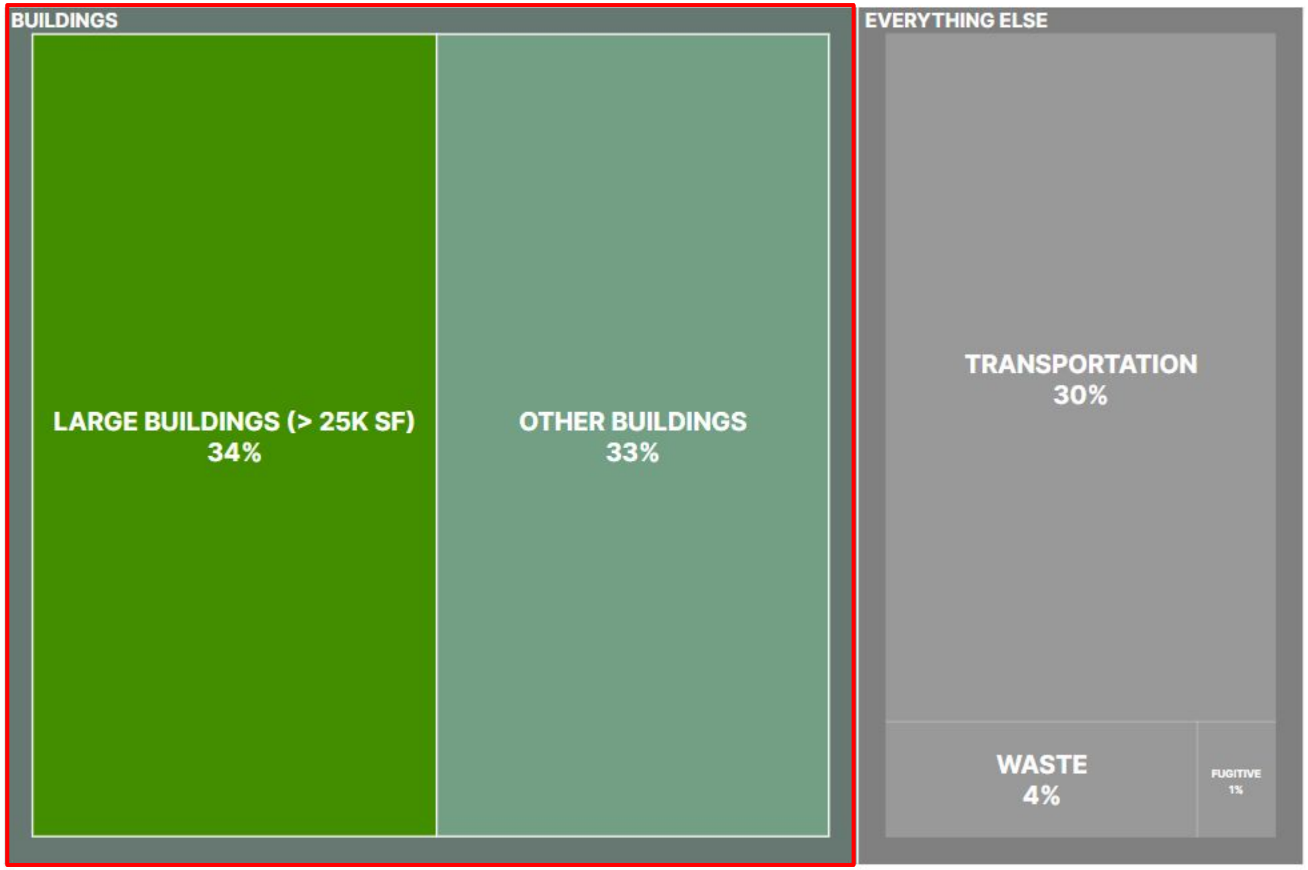
Inspiration



About two thirds of greenhouse gas emissions in NYC come from buildings.

NYC Citywide Greenhouse Gas Emissions by Sector, 2022

Total Emission: 53.7 Million Tonnes CO2e



Data: LL84 2022, NYC Greenhouse Gas Inventory 2022

Most buildings do not meet NYC emissions limits.

Local Law 97, part of New York City's Climate Mobilization Act of 2019, is one of the **country's boldest climate laws**.

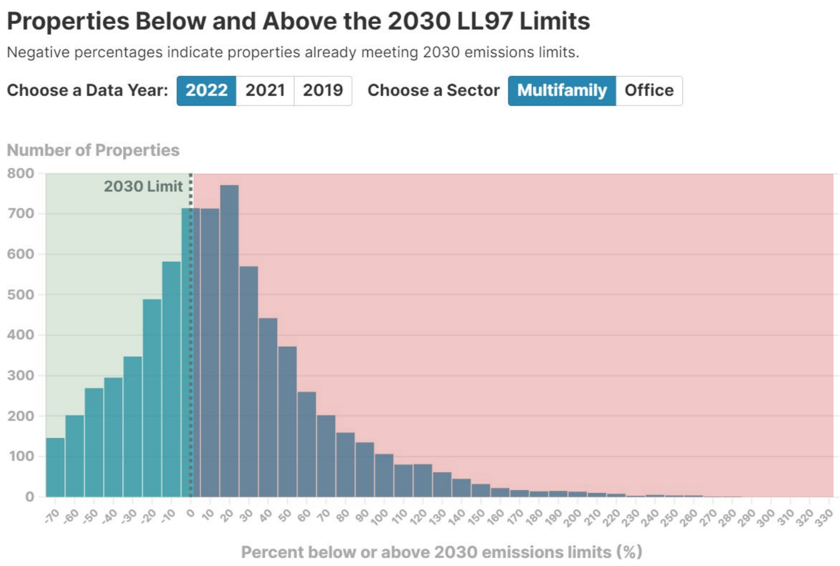
Goal: **cut building carbon emissions by 40% by 2030** and reach **net-zero by 2050** by setting progressively lower caps.

Nearly **two thirds** of all applicable buildings (> 25,000 sq. ft.) **do not meet 2030 emissions limits**.

Several other cities (e.g., Seattle, Boston, and Washington, DC) are enacting similar emissions laws and standards.

65%

Of all buildings in NYC **do not** meet 2030 emissions limits.



The **problem**:

Many buildings **cannot afford the upgrades** needed to meet NYC emissions standards for 2030 and beyond.

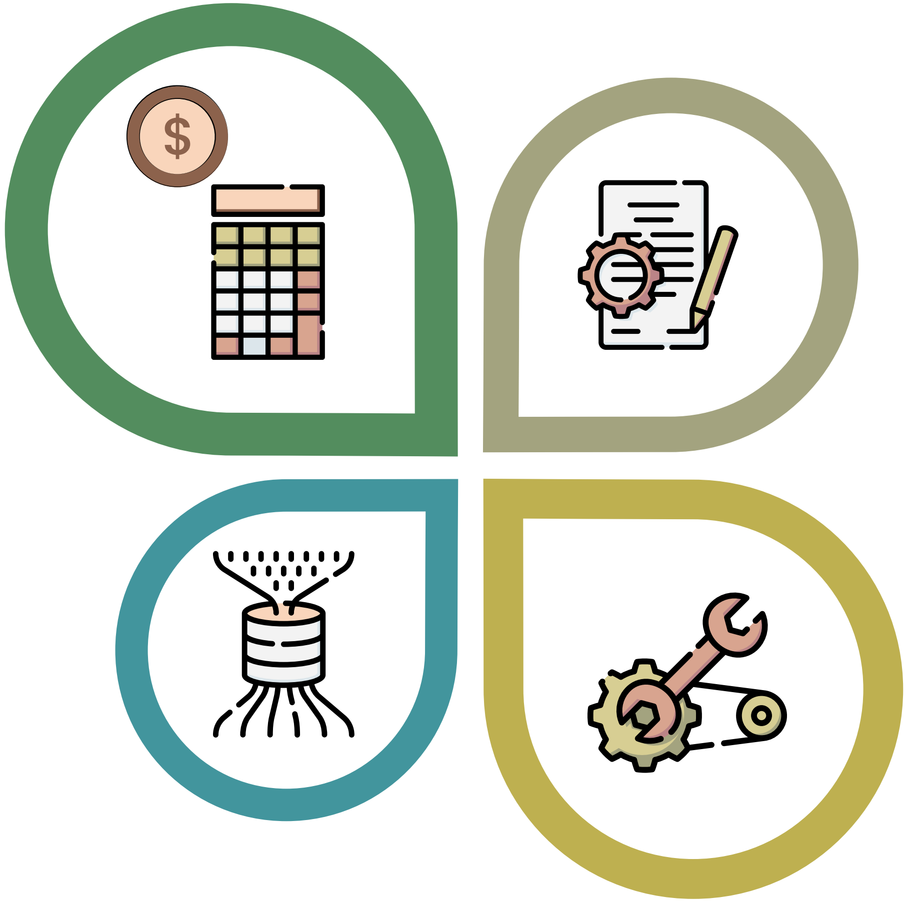
Solution: provide consultative and project management services to help buildings make affordable upgrades to reduce emissions.

Evaluate Building Finances

To determine current spending and budget that they can afford

Assess Building Needs

From data about building emission sources and input from building decision-makers



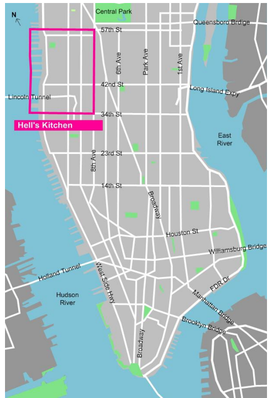
Recommend Upgrades

Based on return on investment from improved efficiency, reductions to emissions, and other secondary benefits

Navigate Implementation

From hiring contractors to applying for rebates, incentives, and financing

Pilot: Multifamily Building in Manhattan

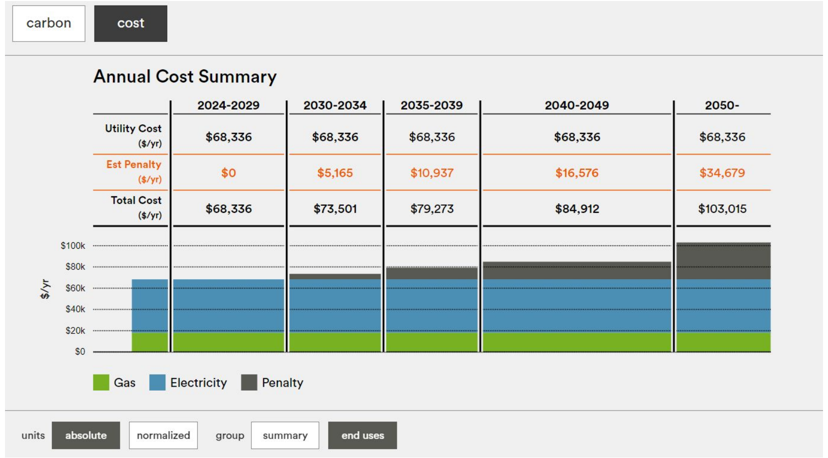


Building-specific needs were also examined and used to guide our initial research

- **Need**: Evaluate the pilot building’s finances and budget for energy efficiency upgrades.
  - **Research Question:** What are the biggest expense categories, and where might there be opportunities to reduce costs?

- **Need**: Improve understanding of the building’s energy consumption and opportunities to lower cost of utilities and LL97 penalties.
  - **Research Question**: How might we define a target reduction of energy consumption for the building?

- **Need**: Complete renovation of aging building roof and façade while staying on-budget.
  - **Research Question:** How might we incorporate energy efficiency improvements into these existing projects?



BE-Ex NYC LL97 Carbon Emissions Calculator

Aging Rooftop of Pilot Building

Several savings targets were identified through an evaluation of building finances

Analysis

- Reviewed building's actual expenditures from 2021 – 2023 financial statements to identify potential cost savings

Findings

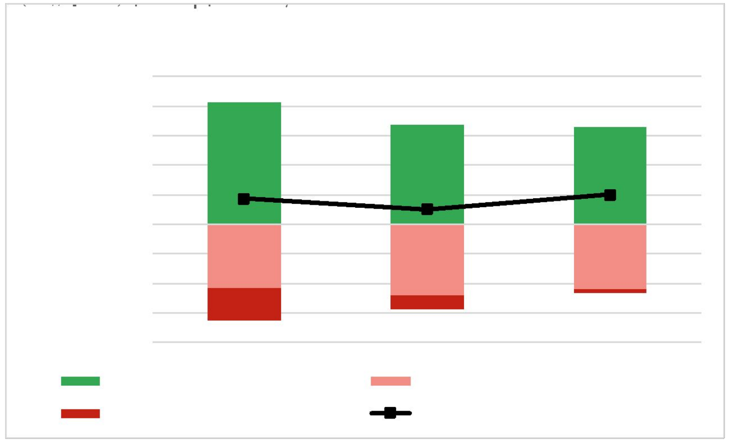
- Average total expenses from 2021 – 2023 were about $225k
- Average grand profit was ~$78k
- Utilities were the biggest expense category, at about $71k (~33% of total).
  - About $32k (46%) was for **gas**
  - About $20k (29%) was for **electricity**
  - About $18k (25%) was for **water/sewer**
- Additional savings might be found in top-ranked expense subcategories

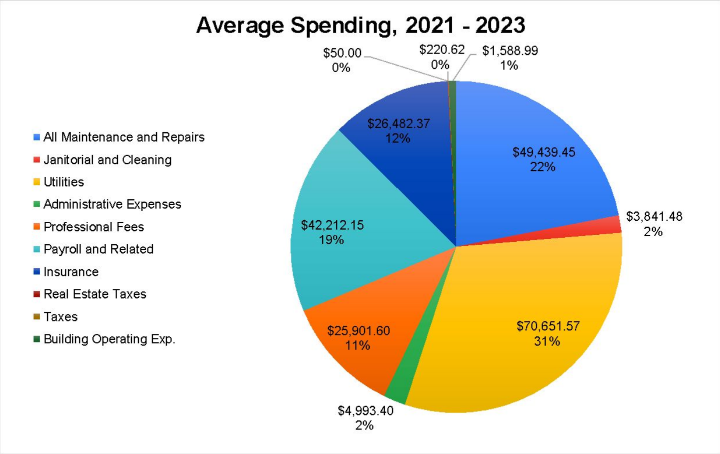
Recommendations

- Request **additional information** from management company about sources of spending in each category/subcategory.
- Seek **competitive pricing** for alternative service providers.

Benefits

- Cost savings could be directed toward more energy efficiency upgrades





710-001 Payroll - Wages	$34,189.93
600-003 Gas	$32,280.85
800-998 Insurance	$20,985.37
600-001 Electricity	$20,404.83
705-002 Management Fees	$18,375.00
600-006 Water & Sewer	$17,965.89
500-001 Elevator Maintenance	$8,519.64
510-004 Plumbing Repairs	$8,256.73
500-005 Exterminator	$5,651.33
705-004 Accounting	$5,125.00
510-019 Heating/Cooling Sys Rep	$5,003.85

Energy bill analysis showed that “Demand” rates contributed significantly to the building’s electricity cost

Analysis

- Averaged gas and electricity consumption and spending for the building’s common areas across the last 16 months

Findings

- Average monthly electricity consumption from building common areas was 6,287 kWh. Average monthly cost of electricity was $1,373.
- Over 45% of the electricity costs came from electricity consumption billed at "demand" rates, but less than 0.2% of the building’s energy consumption was in the “demand” category.
- Per local utility Consolidated Edison (ConEd), if the building consumes less than 5,000 kWh per month, demand rates will be eliminated.

Recommendations

- Reduce electricity consumption by 1,300 kWh / month (~20%)
- Cancel unused gas supply in ancillary apartment (saving 12 – 24 therms and ~$550 / year)
- Perform energy audit to get more details about biggest consumers of electricity within building (e.g., elevator, water pumps, lighting, fans, etc.)

Benefits

- Per LL97 conversion factors, the reduction in annual electricity consumption equivalent to **emissions reductions of about 643.2 tCO2e per year**
- **Cost savings of up to $10,000 per year** could be directed toward more energy efficiency upgrades.

Meter #	New Read	Read Type	Date	Prior Read	Read Type
012782132	7527.89	Actual	Mar 21	7371.07	Actual

Your Supply Charges

Supply 6272.00 kWh @8.516¢/kWh	$534.09
Demand supply 9.96 kW @$15.260/kW	$146.92
Merchant Function Charge	$6.97
GRT & other tax surcharges	$16.56
Sales tax @8.875%	$62.53
Total electricity supply charges	**$767.07**

Date	Read Diff	Multiplier	Total Usage kWh	Total Demand kW
Feb 21	156.81	40	6,272	9.96 - Actual

Your Delivery Charges

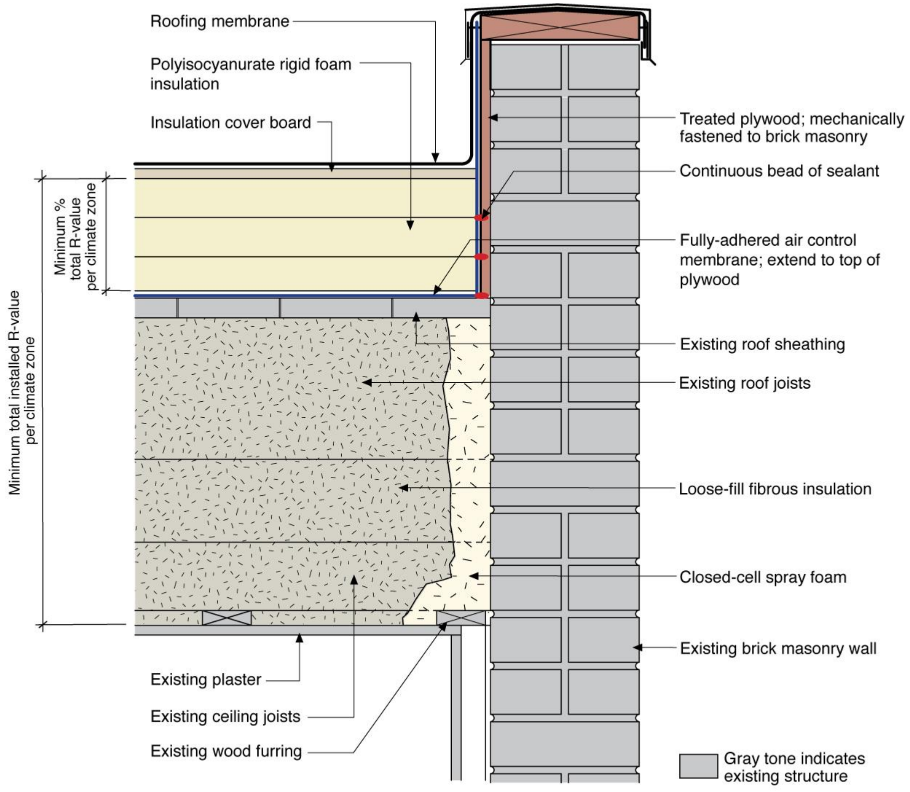
Customer charge	$63.80
Energy delivery 6272.00 kWh @2.965¢/kWh	$185.99
Demand delivery 9.96 kW @ $30.403/kW	$302.81
System Benefit Charge @0.677¢/kWh	$42.48
Billing and payment processing charge	$1.28
GRT & other tax surcharges	$14.36
Sales tax @8.875%	$54.20
Total electricity delivery charges	**$664.92**
Your electricity total	**$1,431.99**

Research on low-slope roofing materials was used to recommend an energy-efficiency approach to planned roof repairs

- **Research**
  - Characterized thermal and reflective **performance of various roofing materials**, and calculated estimated energy and cost savings from their use.
  - Identified **roofing incentives** through the Multifamily Energy Efficiency Program from ConEd; Multifamily Buildings Low-Carbon Pathways Program from New York State Energy Research and Development Authority (NYSERDA); and the NYC Cool Roofs Program.

- **Recommendations**: use phenolic foam, polyiso, or closed-cell polyurethane insulation in combination with a cool roof coating in final roofing project.

- **Benefits**: **up to ~$35k in discounts** from incentive programs, with an **additional $15k discount** if the incentive application package is submitted by June 30, 2024.*



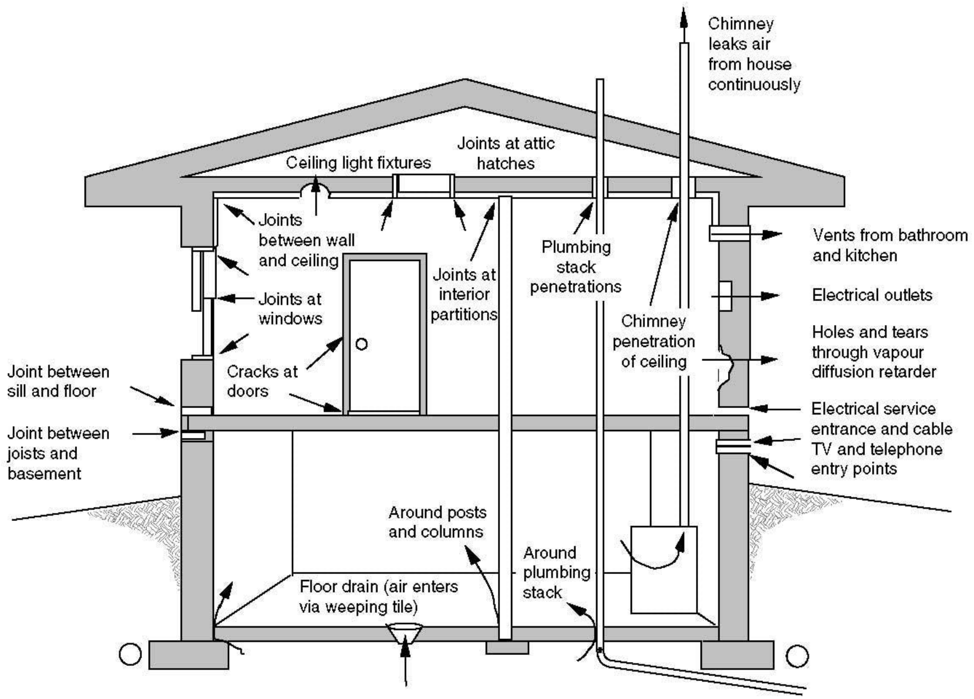
*Calculations of total reductions to energy consumption are pending validation.

Research on additional envelope upgrades was used to recommend add-on work for planned façade repairs

- **Research**
  - Profiled various façade-related upgrades (e.g., air sealing measures, wall insulation, thermal bridge mitigation, high performance windows and doors).
  - Identified additional envelope-related incentives in the ConEd and NYSERDA programs.

- **Recommendations:** perform air sealing around window frames, exterior doors, and louver vents while scaffolding is up and incentives are available.

- **Benefits**: could receive **discounts of up to $1400 (plus an additional $3/therm)** from air sealing incentives.*



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Lighting, lighting controls, and plug loads were also investigated as a solution due to low relative cost

Research

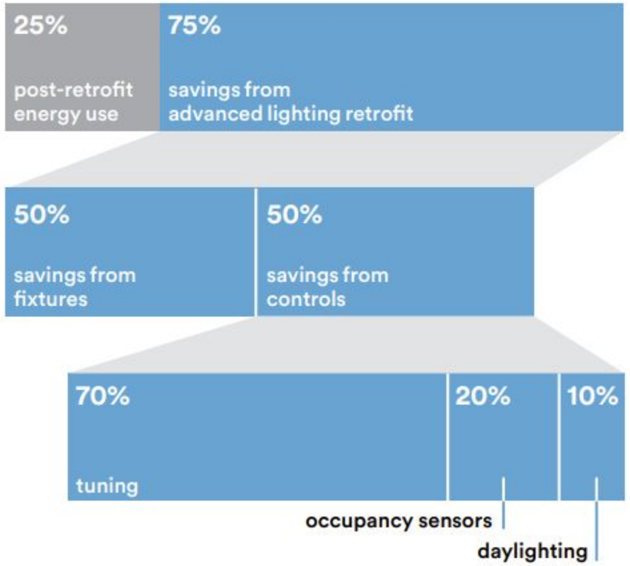
- Profiled various lighting upgrades, such as lamp and ballast replacements, fixture retrofits, and advanced sensors and controls
- Identified LED lighting fixture upgrade incentives through ConEd's Multifamily Energy Efficiency and Instant Lighting Incentives programs.
- Performed building walkthrough with ConEd-designated lighting contractor to quantify impact of upgrades to bi-level LED fixtures throughout building.

Recommendations

- Install bilevel LED fixtures throughout building hallways, stairwells, and other common areas.
- Manage plug loads in ancillary apartment, particularly for appliances (e.g., refrigerator).

Benefits

- **Estimated 80% reduction in lighting-related electricity consumption** compared to existing system.



*Calculations of total reductions to energy consumption are pending validation.

Our next steps are to complete research on HVAC and water heating systems and summarize all findings in a case study

Research Ventilation Systems

- Balanced ventilation and energy recovery systems work in tandem with an airtight envelope to manage air intake and recover heat energy that would otherwise be exhausted to the outdoors, thereby reducing energy use.
- Need to research available ventilation systems and see which might work best for the building.

Research Heating and Cooling Systems

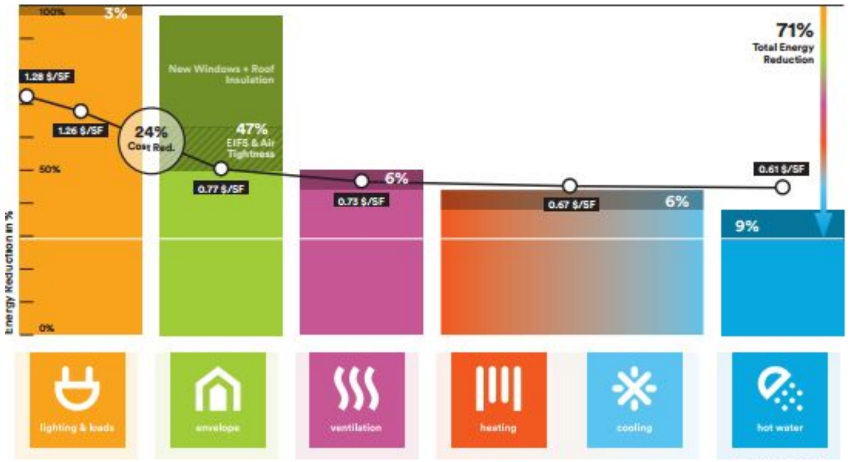
- Heating and cooling is currently provided through an aging natural gas boiler and in-apartment Packaged Terminal Air Conditioners (PTACs)
- Need to research available heating and cooling systems and see which might work best for the building (e.g., Heat Pump PTACs, Ductless Heat Pumps, etc.)
- Since the boiler is owned by the building and the PTACs are owned by the apartment, additional research will be needed to determine cost-sharing strategies
- Project costs have already been estimated based on similar projects.

Research Hot Water Systems

- The building also has an aging natural gas water heater
- Additional research is needed on heat pump water heaters
- The building owns electric water heater units, so it may be worth evaluating a hybrid solution.

Deliverables:

- Updated pitch deck for long-term upgrades for pilot building.
- Case study on pilot building.



BE-Ex Low Carbon Multifamily Retrofit Playbooks: Post-1980 8+ Stories

Aging Boiler in Pilot Building

Aging Water Heater in Pilot Building

Long-term Vision: Expand to Other Buildings and Cities

Background

- Expand the model used for the pilot building to a business focused on consulting and project management services to help buildings afford the full range of building upgrades needed emissions standards

Activities:

- Research the market (buildings in NYC that are likely to encounter challenges affording LL97 upgrades).
- Research potential competitors in the space to identify unoccupied niches and key differentiators.
- Build a business plan that includes a profit model and commercialization strategy.

Deliverables:

- Written business plan.

Impact:

- Emissions reduction the 50,000+ buildings that are held to LL97 emissions standards.

Team-related Needs

- **Financial professionals**, particularly those with experience with navigating building budgets, financing (mortgages, loans, or energy service agreements), or carbon credit purchasing
- **Marketing or sales professionals**, particularly those who have experience with retrofits or upgrades sold to residential buildings (or similar)
- **Architects, Civil Engineers, General Contractors, and other technical professionals** who have experience with technology or materials related to improving building energy efficiency
  - Examples: HVAC systems, water heating systems, low flow water distribution systems, roof and wall insulation, air sealing, windows, lighting, building automation or energy management systems.
- **Project Managers**
- **New York City residents**, particularly those who live in buildings where energy efficiency upgrades are needed, planned, or being implemented
- Any others with a strong interest in the project!

Thanks again to everyone on the team!

(Slightly Edited!) Team Photo at Newlab in Brooklyn Navy Yard